CANONICAL FUNCTORIAL ISOMORPHISM BETWEEN THE DETERMINANTS OF A FINITE LENGTH MODULE OVER A LOCAL RING AND OVER A LOCAL QUOTIENT-RING EXTENSION WITH THE SAME RESIDUE FIELD

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 02PC (sources/stacks-chow/pinned-original-source.tex line 16434).

Allowed standard prerequisites (editorial).

- Length of a finite length module is preserved under restriction along a local ring homomorphism. Algebra, Lemma labelled algebra-lemma-pushdown-module (Stacks tag 02M0), invoked at proof lines 16451-16452. Supplies that M_{R'} has the same finite length l as M, which is what makes the two determinant modules one dimensional over the same residue field.
- The determinant of a finite length module and the isomorphisms gamma for short exact sequences. Same chapter, Lemma labelled lemma-det-exact-sequences (source line 16203), invoked in the statement and at proof line 16462. Supplies the determinant construction and the compatibility assertion whose verification is claimed in the proof; the symbol-wise definition of the isomorphism is inside the counted span.

Source reference. Original lemma 02PA: sources/stacks-chow/pinned-original-source.tex, source lines 16203–16203. Same-chapter pointer into the author environment lemma at source line 16203; invoked at proof line 16462 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02M0: sources/stacks-chow/tags.txt, source lines 3427–3427. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 16452 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $(R', \mathfrak{m}') \to (R, \mathfrak{m})$ be a local ring homomorphism which induces an isomorphism on residue fields κ. Then for every finite length R-module the restriction $M_{R'}$ is a finite length R'-module and there is a canonical isomorphism*

$$\det_{R,\kappa}(M) \longrightarrow \det_{R',\kappa}(M_{R'})$$

This isomorphism is functorial in M and compatible with the isomorphisms $\gamma_{K \to L \to M}$ of Lemma 02PA defined for $\det_{R,\kappa}$ and $\det_{R',\kappa}$.

Proof. If the length of M as an R-module is l, then the length of M as an R'-module (i.e., $M_{R'}$) is l as well, see Algebra, Lemma 02M0. Note that an admissible sequence $x_1, \ldots, x_l$ of M over R is an admissible sequence of M over R' as $\mathfrak{m}'$ maps into $\mathfrak{m}$. The isomorphism is obtained by mapping the symbol $[x_1, \ldots, x_l] \in \det_{R,\kappa}(M)$ to the corresponding symbol $[x_1, \ldots, x_l] \in \det_{R',\kappa}(M)$. It is immediate to verify that this is functorial for isomorphisms and compatible with the isomorphisms γ of Lemma 02PA. $\square$